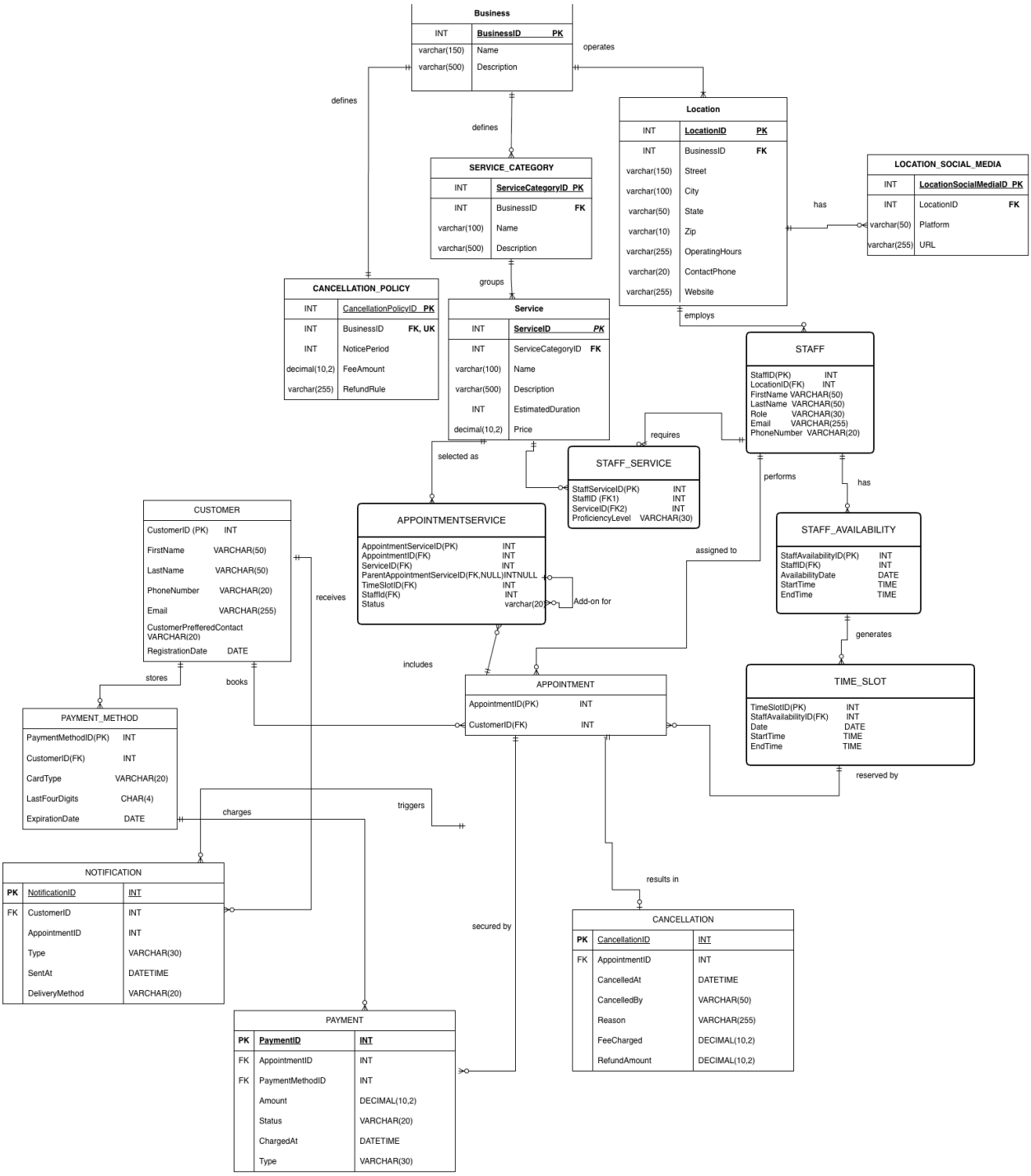


Changes for this phase:

Attribute corrections (addressing “no multivalued, no composite attributes”):

Payment.Status, Payment.Type, Notification.Type, and PaymentMethod.CardType were mistakenly marked multivalued — corrected to ordinary single-valued attributes, each drawing from a fixed set of allowed values rather than holding multiple values at once.

Location.Address (a composite attribute) was split into atomic columns: Street, City, State, Zip.

Location.SocialMedia was genuinely multivalued (a location can have several platform links), so it couldn't just be "un-multivalued" — it became its own entity instead: LocationSocialMedia (LocationSocialMediaID PK, LocationID FK, Platform, URL), one row per link.

Cancellation.Outcome (a vague qualitative label) was replaced with two explicit numeric attributes, FeeCharged and RefundAmount, since the outcome is fully derivable from those two numbers.

Added Customer.CustomerPreferredContact and Location.Website.

Keys and data types (new P3 requirements):

Every entity now has an explicit primary key, and a foreign key for every relationship it participates in.

Every attribute has an assigned data type from a shared vocabulary (INT, VARCHAR(n), DATE, TIME, DATETIME, DECIMAL(10,2), BOOLEAN).

TimeSlot gained a Date attribute (type DATE) to distinguish the same weekday slot across different calendar weeks — needed once waitlisting allowed multiple candidate appointments per slot.

StaffService kept its ProficiencyLevel attribute from the conceptual phase.

Cancellation policy simplified:

Business–CancellationPolicy is now a strict 1:1 (enforced with FK, UK on CancellationPolicy.BusinessID), replacing an earlier idea of policies varying by service type or staff proficiency.

This removed two relationships that only existed to support multiple competing policies: StaffService→CancellationPolicy and a direct Cancellation→CancellationPolicy snapshot link — both unnecessary once there's only one policy per business.

Booking structure reworked (the biggest structural change):

The bare many-to-many between Appointment and Service was resolved into an associative entity, AppointmentService — one row per service selected within one appointment, enabling multi-service appointments and add-ons.

AppointmentService carries a self-referencing FK (ParentAppointmentServiceID, nullable) so an add-on line item can point to the specific core-service line item it belongs to.

Once different staff members could handle different services within one visit, StaffID, TimeSlotID, and Status all moved from Appointment down to AppointmentService — each line item now has its own staff assignment, its own reserved time slot, and its own independent status (so one service in a visit can be confirmed while another is waitlisted).

Appointment itself simplified down to just AppointmentID and CustomerID. EstimatedCharge and ScheduledDateTime were dropped as stored columns — they're now computed values (sum of line-item prices; earliest/latest time slot among the line items) rather than independent facts, which is the correct call for normalization even though it means computing them at query time.